

EE 115 Homework 9

- 1) Consider the phase locked loop (PLL) with the loop gain $K = \frac{1}{2}A_u A_v K_v$ and the loop filter $G(s)$. The input to the PLL is $u(t) = A_u \cos(2\pi f_c t + \theta_i(t))$, the output of the voltage controlled oscillator (VCO) in the PLL is $v(t) = -A_v \sin(2\pi f_c t + \theta_o(t))$, and

$$\theta_o(t) = K_v \int_0^t x(\tau) d\tau \quad (1)$$

with $x(t)$ being the output of the loop filter $G(s)$. (Please use the above definition of $x(t)$.)

If the phase error $\theta_e(t) \doteq \theta_i(t) - \theta_o(t)$ is small, it is known that

$$H(s) \doteq \frac{\Theta_o(s)}{\Theta_i(s)} = \frac{KG(s)}{s + KG(s)}. \quad (2)$$

- a) **(30 points)** Assume $G(s) = 1$ and $\theta_i(t) = \Delta\theta u(t)$. Determine and sketch $\theta_o(t)$. What is $\theta_o(\infty)$?
- b) **(30 points)** Assume $G(s) = 1$ and $\frac{d\theta_i(t)}{dt} = 2\pi k_f u(t)$. Determine and sketch $\frac{d\theta_o(t)}{dt}$, $x(t)$ and $\theta_e(t)$. How does $\theta_e(\infty)$ depend on K ?
- c) **(40 points)** Assume $G(s) = \frac{s+a}{s}$ and $\frac{d\theta_i(t)}{dt} = 2\pi k_f u(t)$. Determine $x(\infty)$ and $\theta_e(\infty)$.